Make the evidence chain visible.

KAPPA

From medicinal plant context to testable molecular hypotheses.

An evidence-aware research workbench for natural-product discovery—not a treatment recommender.

PLANT
Curcuma longa

MOLECULE
Curcumin

BIOACTIVITY
Human PTGS2

MODEL
Scaffold-held-out ML

ETHNOHACK 2026 · NATURAL PRODUCTS FOR DRUG DISCOVERY

The challenge is disconnected evidence.

Botanical records, compound structures, target assays and human genetics live in different systems—with different identifiers, evidence standards and reuse rules.

Occurrence

Where botanical evidence appears

Chemistry

What structure the identifier means

Bioactivity

Which assay measured what

Pharmacogenomics

What variants may or may not support

A plant occurrence is not a medicinal-use claim. A biochemical activity is not clinical efficacy.

KAPPA makes those boundaries explicit while creating a reproducible path between datasets.

Reproducible hand-offs, not another mega-database.

LOTUS, IMPPAT, ChEMBL, Open Targets and NP-KG already provide valuable data and integration capabilities.

KAPPA’s bounded contribution: connect a traceable occurrence context to a measured target-specific dataset, a scaffold-aware baseline, an applicability check, and evidence-tiered interpretation.

01 **Live, attributable assay acquisition**

02 **RDKit features + duplicate controls**

03 **Scaffold-held-out RF vs baseline**

04 **Evidence-separated graph + applicability gate**

05 **Interactive research workbench**

Integration/reproducibility work—not a claim to be the first AI, knowledge graph or medicinal-plant database.

One pipeline. Provenance at every hand-off.



Observed assay records → computed molecular features → held-out predictions

Wikidata occurrence and one secondary ethnobotanical report are distinct plant-level evidence. A verified CYP2C19–clopidogrel guideline is a disconnected PGx demo; no link to curcumin/PTGS2.

Real data. Narrow, predeclared labels.

RAW RECORDS

7,482

ChEMBL CHEMBL230 · human PTGS2

CURATED

3,164

canonical structures · duplicate-collapsed

ACTIVE BAND

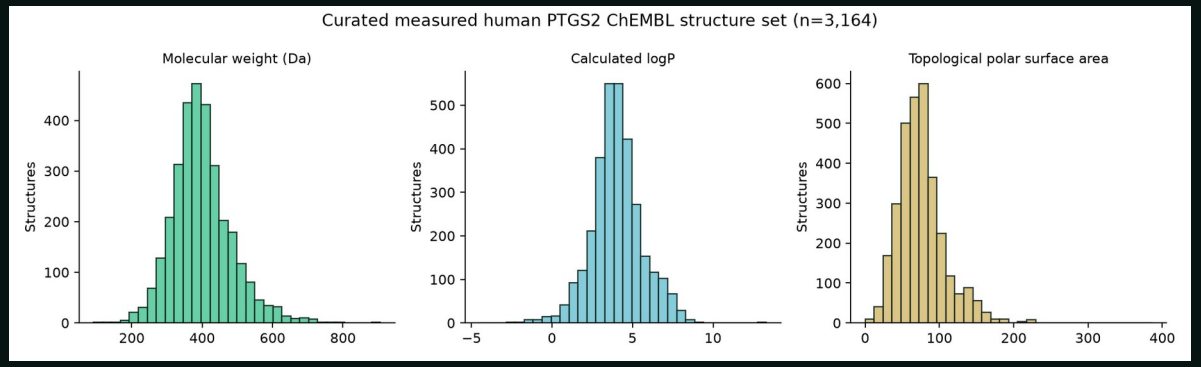
2,505

IC50 ≤ 1 µM

INACTIVE BAND

659

IC50 ≥ 10 µM

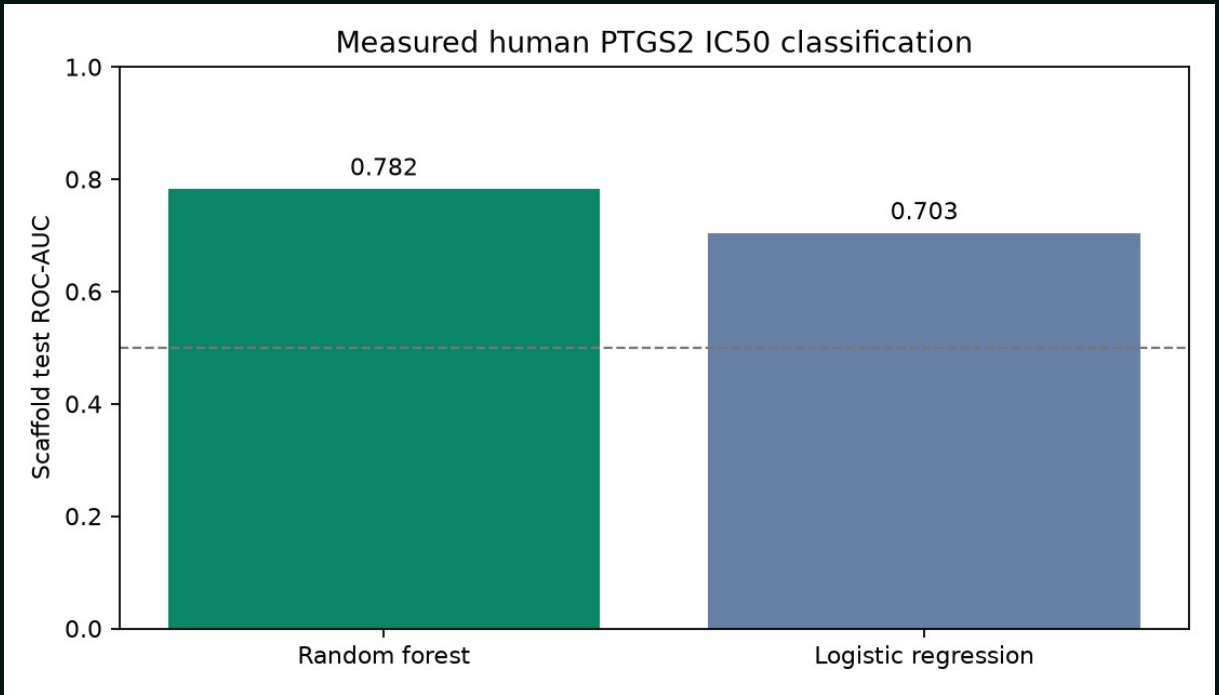


STRICT FILTERS

Exact standard relation · nM units · single-protein assay · Homo sapiens · no duplicate/validity flags.

1–10 µM grey band omitted—not relabelled inactive.

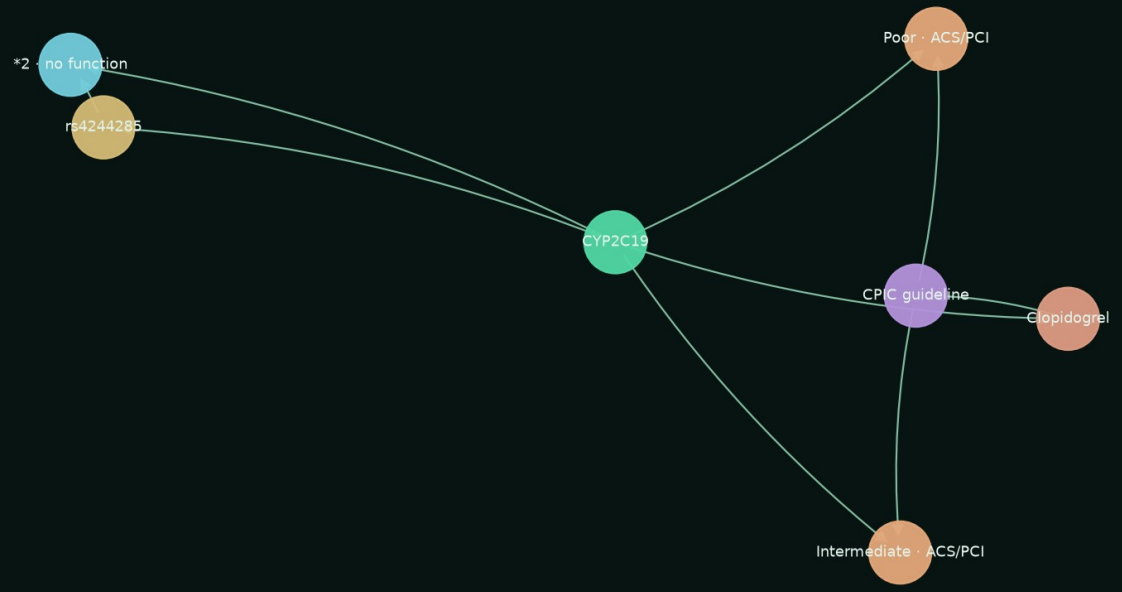
A genuine benchmark—with a visible caveat.



Random Forest	**ROC-AUC 0.782 · PR-AUC 0.940**
Logistic baseline	**ROC-AUC 0.703 · PR-AUC 0.907**
3-fold scaffold CV	**Mean ROC 0.810 · PR 0.935**

Test prevalence 82.6% (PR baseline); balanced accuracy 0.637. Strong imbalance; internal data only, not external validation.

A graph that distinguishes identifiers from inference.



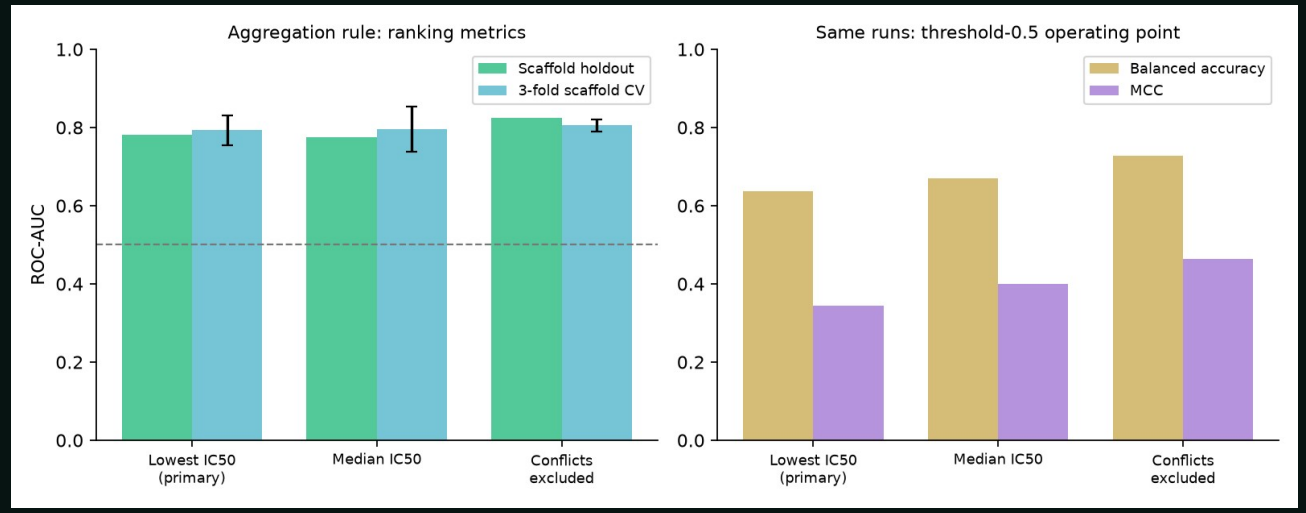
Pharmacogenomic example: verified, contextual, separate.

CYP2C19 *2 and clopidogrel are tied to CPIC guideline 100411 (CPIC A / ClinPGx 1A), with phenotype- and ACS/PCI-conditioned records only.

No patient genotype and no link to curcumin or PTGS2. ClinVar supplies variant identity only: its “Likely benign; other” label covers an unspecified disease trait, not drug-metabolizer function.

Missing evidence ≠ negative biology.

No composite ranking. Keep the prediction in scope.



NOT CLAIMED

- No blended evidence score or evidence-axis weighting
- No calibrated probability and no validated domain boundary
- No external, prospective, wet-lab or clinical validation
- No individual treatment or pharmacogenomic prediction

CURCUMIN · RETROSPECTIVE

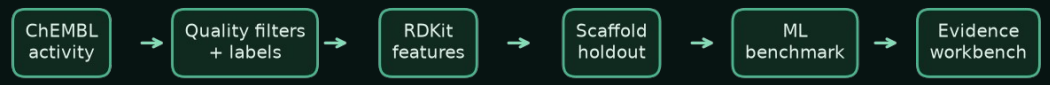
Exact training-structure overlap

Uncalibrated model-only output

Applicability cutoff 0.324 is a training-only similarity diagnostic—not a calibrated confidence guarantee. No evidence-axis average.

Duplicate rule matters: 32 structures conflict. ROC 0.774–0.825, balanced accuracy 0.637–0.729 — ranking stable, threshold operating point not. Hyperparameter and time/document tests are internal ChEMBL, not external validation.

A working workbench—not a screenshot.



Explore	Measured structures and descriptors
Inspect	Activity bands, assays and source IDs
Predict	SMILES + similarity + overlap flag
Traverse	Typed graph with evidence notes
Export	Curated CSV and reproducible metrics

Live demo: plant context → compound table → model card → graph → methods/provenance.

Useful now. Clear about what comes next.

DELIVERED

Working API + UI (local)

3,164-record data pipeline

Scaffold benchmark + 15 tests

Evidence graph + docs

10-slide deck + rendered PDF

NEXT VALIDATION

Primary-source ethnobotanical evidence + permission

Independent assay/database evaluation

Clinical PGx update + indication checks

Probability calibration and domain review

Prospective experimental follow-up

KAPPA turns fragmented evidence into a traceable research question—not a therapeutic promise.